

AFZAL AMANULLAH

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EDUCATION

Bachelor of Science (B.S) - Data Science and Application
Indian Institute of Technology, Madras, Tamil Nadu | Currently pursuing

2025 – 2029 (Expected)

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C, SQL, HTML5, CSS3

Frameworks & Libraries: Rasterio, GDAL, Matplotlib, Streamlit, Androguard, React.js, Node.js, Flask, Flask-SocketIO, Tailwind CSS, Vite, React Router, Leaflet.js

AI, Machine Learning & Computer Vision: XGBoost, Computer Vision, Deep Learning, Scikit-Learn, PyTorch, YOLOv8, OpenCV, Gemini 2.5 Flash SDK, Claude/Anthropic API, OpenRouter, Prompt Engineering, Spatial Forecasting

Data & Backend: Relational Database, Pandas, NumPy, Sparse Matrix Vectorization, Firebase, Firestore, REST APIs, Geolocation APIs, Role-Based Access Control, SQLite

Cybersecurity: Network Intrusion Detection, Ethical Hacking, Vulnerability Assessment, Secure Coding Practices, Packet Analysis (Scapy) Hardware & IoT: GPU Acceleration (CUDA), ESP32, Arduino, Serial Communication, Sensor Integration

Engineering Practices: Data Structure & Algorithm (DSA), Mixed Precision (AMP), SDLC, Agile Methodology, System Architecture, API Design, Git/Version Control, Anomaly Detection, Feature Engineering, Mathematical Modeling

PROJECTS & APPLIED EXPERIENCES

Deep Learning Based Super Resolution Satellite Image Mapping

Aug 2026

[\[Kaggle Notebook\]](#)

- Engineered a 4x Super-Resolution framework using a ResNet and PixelShuffle to upscale 10m Sentinel-2 imagery to sub-4m resolution.
- Quantified epistemic uncertainty by integrating Monte Carlo Dropout to generate pixel-level confidence heatmaps for geospatial analysis.
- Optimized training pipelines using Rasterio windowed reads, FP16 mixed precision, and OneCycleLR scheduling, achieving a validation PSNR of 23.91 dB.

Hybrid AI Automated Malware Detection Engine

Jun 2026

[\[Kaggle Notebook\]](#)

- Architected a hybrid ML and Generative AI pipeline to stream and analyze real-world APK binaries from the AndroZoo repository.
- Automated reverse-engineering via Androguard, piping semantic profiles into Gemini 2.5 Flash for explainable threat verdicts.
- Processed 27.3M records (12GB) using under 50MB of RAM via O(1) hash lookups, deploying a Streamlit dashboard that calculates a 60/40 weighted composite threat score.

AI/ML Financial Fraud & Mule Account Detection System | [\[Github Repository\]](#)

Jul 2026

- Built a high-dimensional fraud detection pipeline using anonymized real-world transaction data provided by **Bank of India (BOI)**, processing ~4,000 sparse features while applying strict filters to prevent data leakage.
- Developed a GPU-accelerated XGBoost classifier with dynamic scaling weights to resolve severe class imbalances between legitimate and fraudulent accounts.
- Achieved a 93.0% ROC-AUC score by applying rigorous defensive hyperparameter regularization to prevent model overfitting

Real-Time Air Quality Prediction & IoT Forecasting System | [\[Github Repository\]](#)

Mar 2026

- Engineered a spatial forecasting platform using Advection-Diffusion Partial Differential Equation (PDE) models to predict pollution spread 24 hours in advance.
- Developed a low-cost mesh IoT prototype (Arduino UNO, MQ-135, HC-05) that dynamically integrates live meteorological REST API data.
- Designed an interactive forecasting dashboard featuring a custom "Pollution Urgency Score" that mathematically factors in wind dispersion and rain scavenging.

LEADERSHIP & CAMPUS INVOLVEMENT

Technical Team Lead

2025 – Present

- Captained Team Mango in developing enterprise-grade banking security prototypes, directing the architecture of a GenAI-powered automated malware analysis system for fraudulent APKs and an AI/ML classification engine for suspicious mule account detection
- Captained cross-functional engineering squads for the Smart India Hackathon 2026, IChE-ChEMATHON Grand Finale, and e-Yantra (eLSI), directly managing AI architecture, task delegation, and prototype deployment.